

Feedback — Programming Homework 1

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You submitted this homework on **Sun 12 Jul 2015 1:03 PM PDT**. You got a score of **6.00** out of **7.00**.

In lecture and in a practical, we saw an implementation of the naive exact matching algorithm:

```
def naive(p, t):
    occurrences = []
    for i in range(len(t) - len(p) + 1): # loop over alignments
        match = True
        for j in range(len(p)): # loop over characters
            if t[i+j] != p[j]: # compare characters
                match = False
                break
        if match:
            occurrences.append(i) # all chars matched; record
    return occurrences
```

...and we saw a function that takes a DNA string and returns its reverse complement:

```
def reverseComplement(s):
    complement = {'A': 'T', 'C': 'G', 'G': 'C', 'T': 'A', 'N': 'N'}
    t = ''
    for base in s:
        t = complement[base] + t
    return t
```

...and we saw a function that parses a DNA reference genome from a file in the FASTA format.

```
def readGenome(filename):
    genome = ''
    with open(filename, 'r') as f:
        for line in f:
            # ignore header line with genome information
            if not line[0] == '>':
                genome += line.rstrip()
    return genome
```

...and we saw a function that parses the read and quality strings from a FASTQ file containing sequencing reads.

```
def readFastq(filename):
```

```
sequences = []
qualities = []
with open(filename) as fh:
    while True:
        fh.readline() # skip name line
        seq = fh.readline().rstrip() # read base sequence
        fh.readline() # skip placeholder line
        qual = fh.readline().rstrip() # base quality line
        if len(seq) == 0:
            break
        sequences.append(seq)
        qualities.append(qual)
return sequences, qualities
```

First, implement a version of the naive exact matching algorithm that is *strand-aware*. That is, instead of looking only for occurrences of P in T, additionally look for occurrences of the *reverse complement* of P in T. If P is ACT, your function should find occurrences of both ACT and its reverse complement AGT in T.

If P and its reverse complement are identical (i.e. P is a *palindrome*), then a given match offset should be reported only once. So if your new function is called `naive_with_rc`, then the old `naive` function and your new `naive_with_rc` function should return the same results when P is a palindrome.

Next, download and parse the lambda virus genome, at:

https://d28rh4a8wq0iu5.cloudfront.net/ads1/data/lambda_virus.fa

Question 1

How many times does AGGT or its reverse complement occur in the lambda virus genome?

You entered:

306

Your Answer	Score	Explanation
306	✓ 1.00	
Total	1.00 / 1.00	

Question 2

How many times does TTAA or its reverse complement occur in the lambda virus genome?

You entered:

195

Your Answer		Score	Explanation
195	✓	1.00	
Total		1.00 / 1.00	

Question 3

What is the offset of the leftmost occurrence of ACTAAGT or its reverse complement in the Lambda virus genome?

You entered:

26028

Your Answer		Score	Explanation
26028	✓	1.00	
Total		1.00 / 1.00	

Question 4

What is the offset of the leftmost occurrence of AGTCGA or its reverse complement in the Lambda virus genome?

You entered:

450

Your Answer		Score	Explanation
450	✓	1.00	
Total		1.00 / 1.00	

Question 5

As we will discuss in more depth, sometimes we would like to find approximate matches for P in T . That is, we want to find occurrences with one or more differences.

For Questions 5 and 6, make a new version of the naive function (not the strand-aware version you made before) called `naive_2mm` that allows up to 2 mismatches per occurrence. For example,

ACTTTA occurs twice in ACTTACTTGATAAAGT, once at offset 0 with 2 mismatches, and once at offset 4 with 1 mismatch. So `naive_2mm('ACTTTA', 'ACTTACTTGATAAAGT')` should return the list `[0, 4]`.

How many times does TTCAAGCC occur in the Lambda virus genome when allowing up to 2 mismatches?

You entered:

Your Answer		Score	Explanation
191	✓	1.00	
Total		1.00 / 1.00	

Question 6

What is the offset of the leftmost occurrence of AGGAGGTT in the Lambda virus genome when allowing up to 2 mismatches?

You entered:

Your Answer		Score	Explanation
12	✗	0.00	

Total

0.00 / 1.00

Question 7

Finally, download and parse the provided FASTQ file containing real DNA sequencing reads derived from a human:

https://d28rh4a8wq0iu5.cloudfront.net/ads1/data/ERR037900_1.first1000.fastq

The reads are taken from this study:

Ajay, S. S., Parker, S. C., Abaan, H. O., Fajardo, K. V. F., & Margulies, E. H. (2011). Accurate and comprehensive sequencing of personal genomes. *Genome research*, 21(9), 1498-1505.

This dataset has something wrong with it; one of the sequencing cycles is poor quality.

Report which sequencing cycle has the problem. For example, if the leftmost read position has the problem, report **0**. If the fourth position from the left has the problem, report **3**. Do whatever analysis you think is needed to identify the bad cycle. It might help to review the practical session where we analyzed reads by position.

You entered:

Your Answer		Score	Explanation
66	✓	1.00	
Total		1.00 / 1.00	